KIT - Kalaignarkarunanidhi Institute of Technology

(An Autonomous Institute)

Department of Computer Applications

M23CAP401– PROJECT

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PROJECT TITLE:

Smart Answering System using Retrieval-Augmented Generation (RAG)

ABSTRACT

- Developed a Smart Answering System using Retrieval-Augmented Generation (RAG).
- Understands user questions in natural language.
- Retrieves relevant information from uploaded documents before generating answers.
- Converts documents into vectors for efficient semantic search.
- Stores vectors in a vector database for fast information retrieval.
- Uses an open-source Large Language Model (LLM) to generate context-based responses.
- Reduces hallucination by grounding answers in retrieved documents.
- Handles large volumes of data efficiently.
- Suitable for chatbots, document search systems, and knowledge management applications.

EXISTING SYSTEM

- Uses only keyword-based search without semantic understanding.
- Primarily works with plain text and struggles to process PDFs or images effectively.
- Stores data in traditional databases without contextual understanding.
- Generates answers without validating information, leading to possible incorrect responses.
- Cannot efficiently handle large unstructured data such as scanned documents.
- Performance decreases as data size increases.
- Requires manual rule-based methods for information extraction.
- Provides less accurate answers for complex or multi-part queries.
- Not suitable for real-time intelligent applications.
- Does not utilize modern AI techniques or multimodal models.

PROPOSED SYSTEM

- Can understand text, images, PDFs, and scanned documents in one system
- Supports different file formats and adjusts automatically
- Uses open-source AI models with a smart agent to understand user questions
- Converts all data into vector form so the system can understand meaning
- Stores the data in a vector database for quick access
- Smart agent analyzes and improves the user question if needed
- Finds the most relevant information based on the question
- Thinks step-by-step to generate accurate answers
- Gives clear, explainable, and context-based responses
- Useful for chatbots, research, document analysis, and decision making

SYSTEM MODULES

- Modules 1- Document Ingestion & Management Module
- Modules 2- Multimodal Processing & Embedding Module
- Modules 3- Vector Storage & Retrieval Module
- Modules 4- Agentic Query Understanding Module
- Modules 5 - Context Reasoning & Answer Generation Module

Module description

Modules 1-document ingestion & management

- Uploads pdfs, images, and text files
- Extracts and organizes document content
- Maintains document metadata for structured processing

Modules 2-multimodal processing & embedding

- Processes text and image-based content and converts it into numerical vectors (embeddings).
- Enables semantic understanding for better comparison and similarity search.

Modules 3 -vector storage & retrieval

- Stores generated embeddings in a vector database for efficient searching.
- Retrieves the most relevant information based on the semantic similarity of user queries

Modules 4-Agentic Query Understanding

- Analyzes user intent and rewrites complex or unclear queries if necessary.
- Plans intelligent retrieval steps to improve answer accuracy.

Moduels 5-Context Reasoning & Answer Generation

- Combines retrieved information with the user query to generate structured and context based responses.
- Ensures answers are meaningful, accurate, and grounded in the uploaded documents.

MODELS

- **Large Language Model (LLM):**

  Open AI

- **Embedding Model:**

  Sentence Transformers (all-MiniLM-L6-v2)

- **Vision Model:**

  CLIP (Contrastive Language–Image Pretraining)

- **OCR Model:**

  Tesseract OCR / EasyOCR

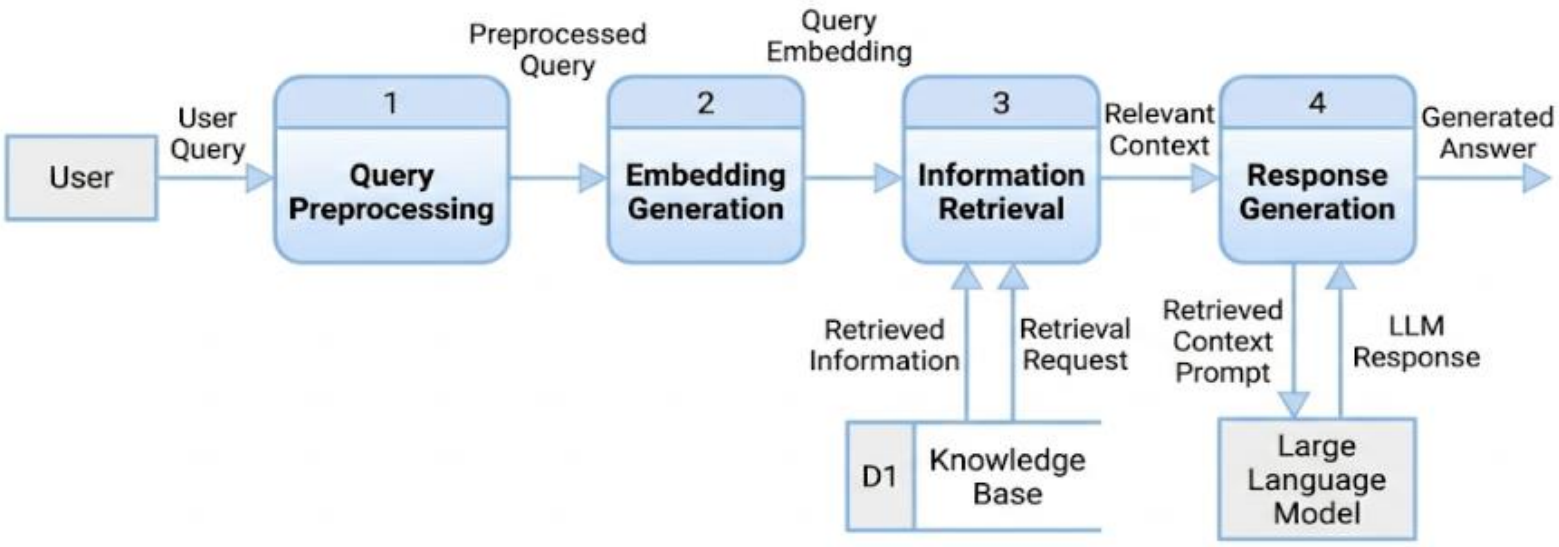
SYSTEM SPECIFICATION

- **HARDWARE**
- Processor: Intel Core i5 or higher
- RAM: Minimum 8 GB (16 GB recommended)
- Storage: Minimum 256 GB HDD/SSD
- System Type: 64-bit computer

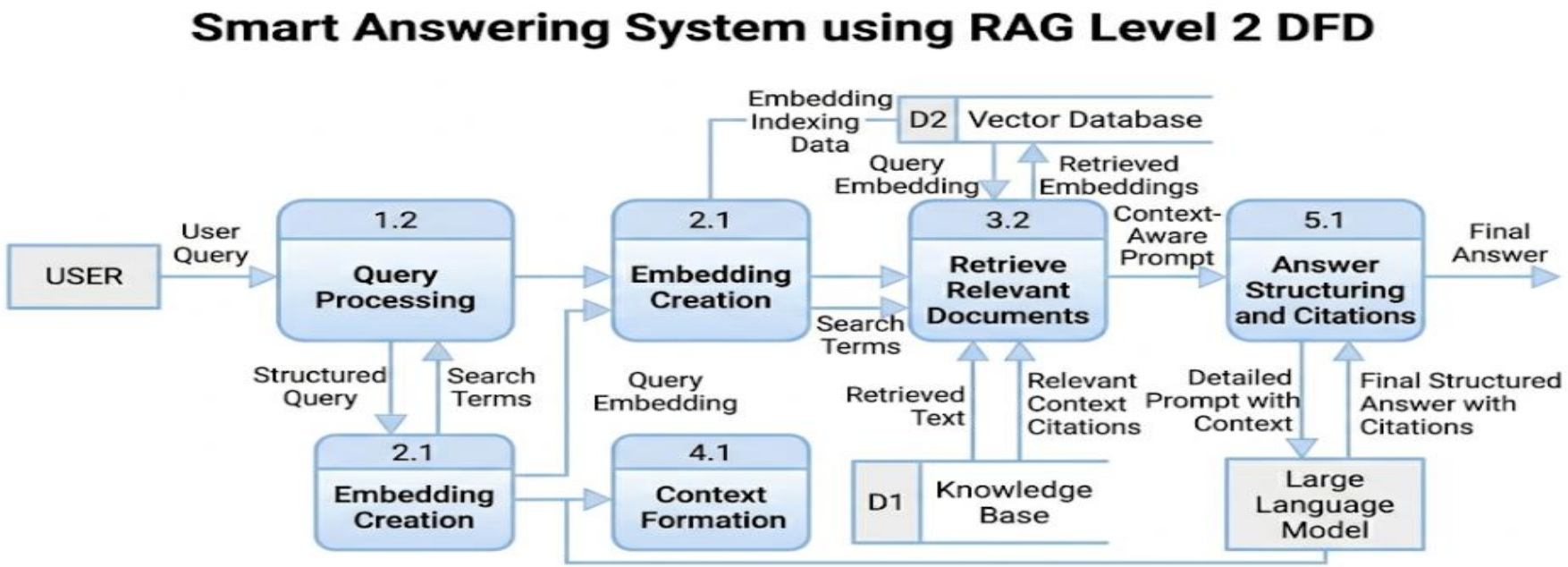
- **SOFTWARE**
- Operating System: Windows 11, Linux, or macOS
- Programming Language: Python 3.10 or above
- Database: Chroma Vector Database
- Tools: VS Code

DATA FLOW DIAGRAM

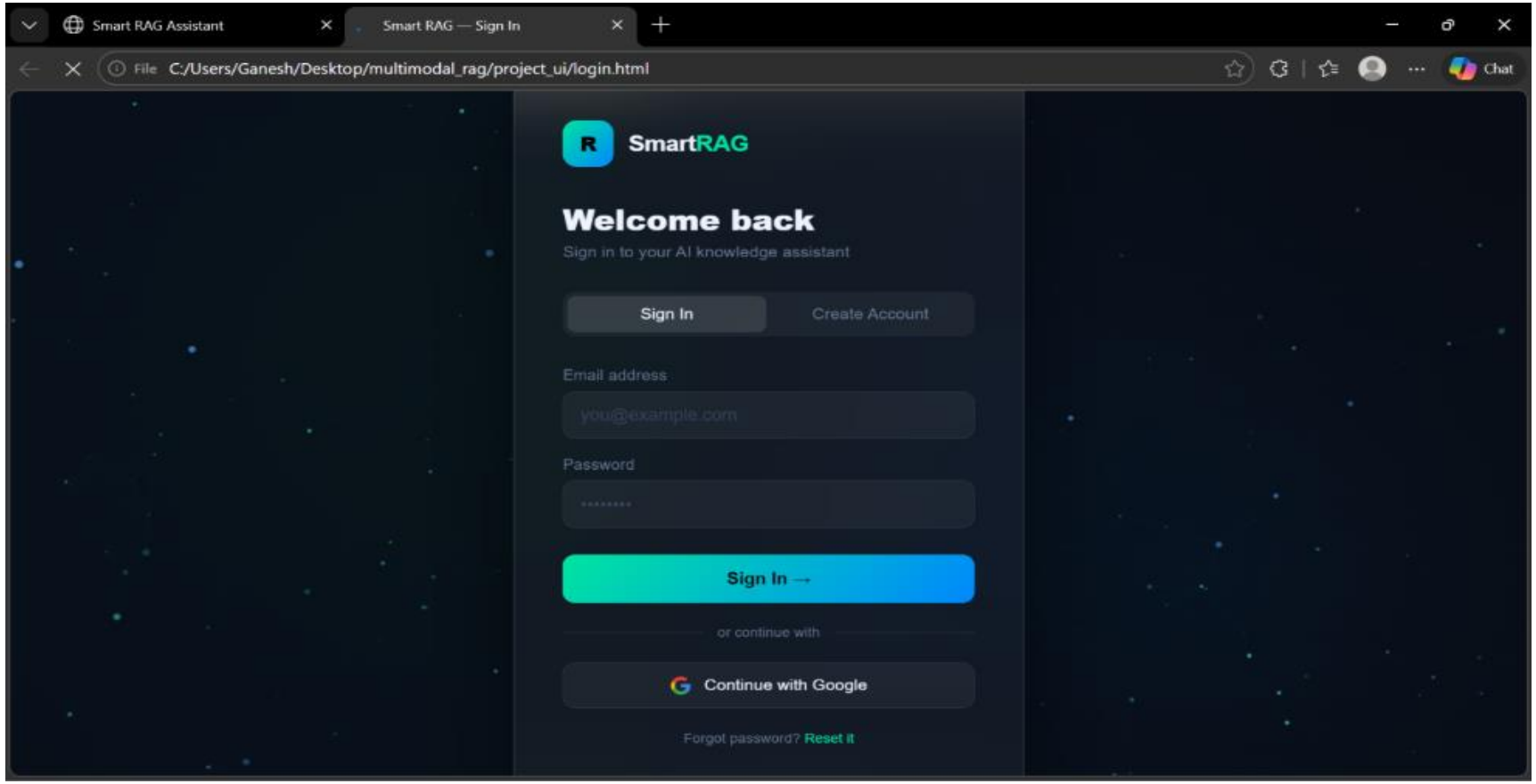
Level 1

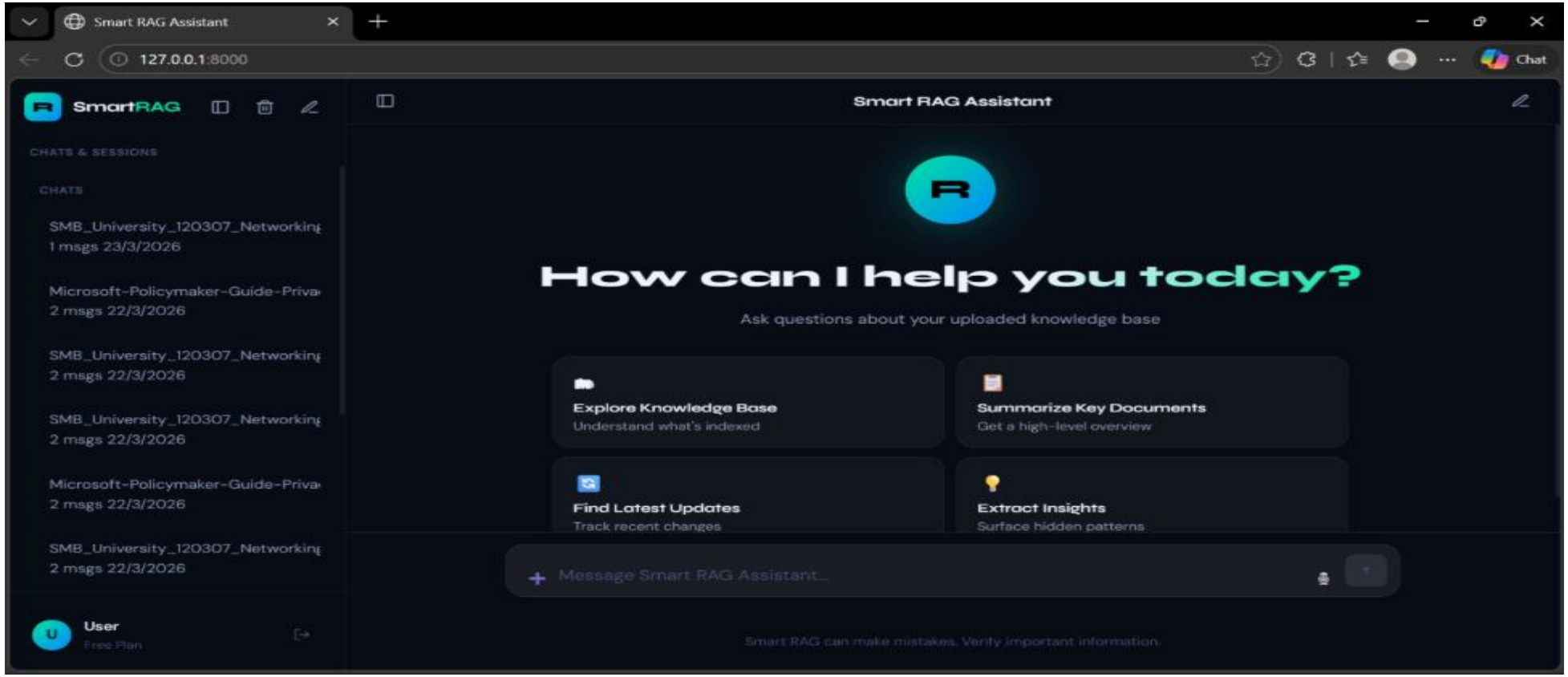


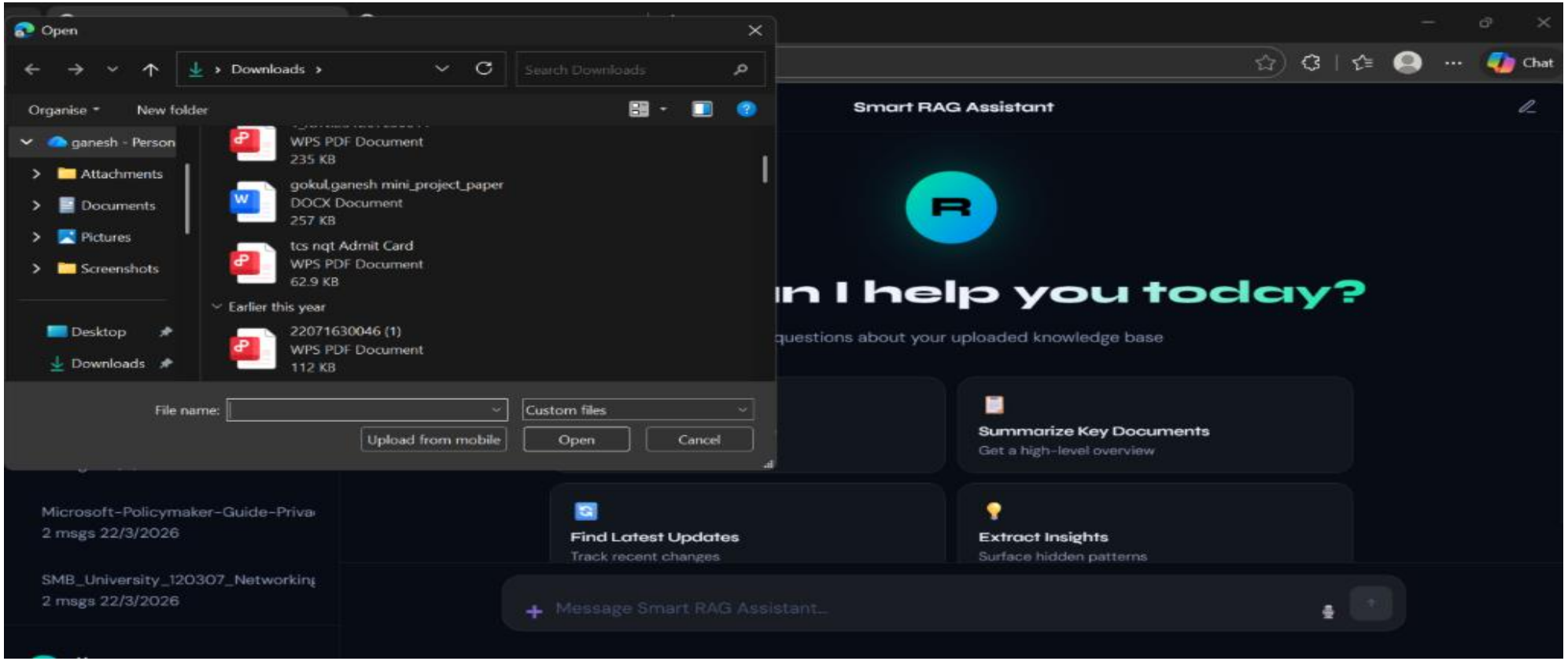
Level 2

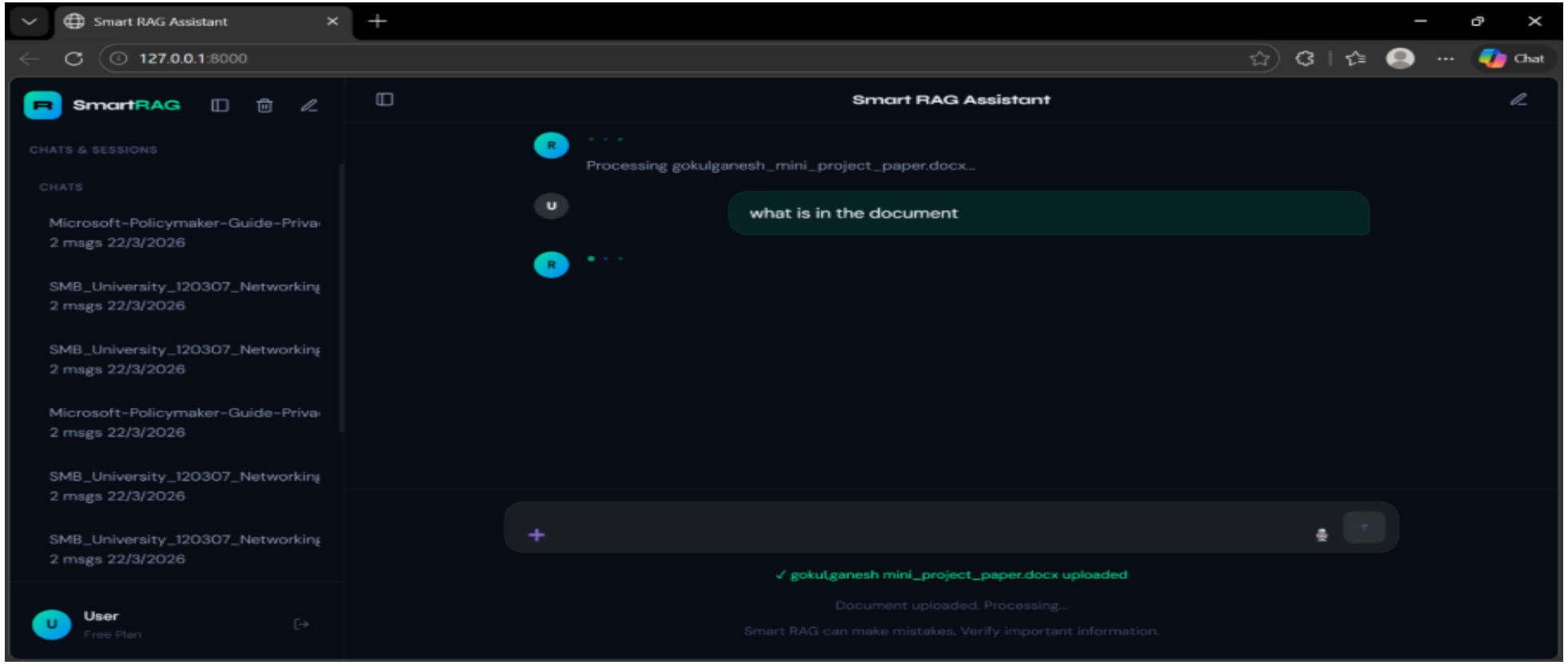

Smart Answering System using RAG Level 2 DFD
USER
User Query
1.2
Query Processing
Structured Query
Search Terms
2.1
Embedding Creation
Embedding Indexing Data
D2
Vector Database
Query Embedding
Retrieved Embeddings
2.1
Embedding Creation
Search Terms
3.2
Retrieve Relevant Documents
Context-Aware Prompt
5.1
Answer Structuring and Citations
Final Answer
Query Embedding
4.1
Context Formation
Retrieved Text
Relevant Context Citations
D1
Knowledge Base
Detailed Prompt with Context
Final Structured Answer with Citations
Large Language Model

screenshot


Smart RAG Assistant
Smart RAG — Sign In
File C:/Users/Ganesh/Desktop/multimodal_rag/project_ui/login.html
Chat
SmartRAG
Welcome back
Sign in to your AI knowledge assistant
Sign In
Create Account
Email address
you@example.com
Password
Sign In →
or continue with
Continue with Google
Forgot password? Reset it


Smart RAG Assistant
127.0.0.1:8000
Chat
SmartRAG
Smart RAG Assistant
CHATS & SESSIONS
CHATS
SMB_University_120307_Networking
1 msgs 23/3/2026
Microsoft-Policymaker-Guide-Priva
2 msgs 22/3/2026
SMB_University_120307_Networking
2 msgs 22/3/2026
SMB_University_120307_Networking
2 msgs 22/3/2026
Microsoft-Policymaker-Guide-Priva
2 msgs 22/3/2026
SMB_University_120307_Networking
2 msgs 22/3/2026
How can I help you today?
Ask questions about your uploaded knowledge base
Explore Knowledge Base
Understand what's indexed
Summarize Key Documents
Get a high-level overview
Find Latest Updates
Track recent changes
Extract Insights
Surface hidden patterns
Message Smart RAG Assistant...
User
Free Plan
Smart RAG can make mistakes. Verify important information.


Open
Downloads
Search Downloads
Organise
New folder
ganesh - Person
Attachments
Documents
Pictures
Screenshots
Desktop
Downloads
WPS PDF Document
235 KB
gokul.ganesh mini_project_paper
DOCX Document
257 KB
tcs nqt Admit Card
WPS PDF Document
62.9 KB
Earlier this year
22071630046 (1)
WPS PDF Document
112 KB
File name:
Custom files
Upload from mobile
Open
Cancel
Chat
Smart RAG Assistant
n I help you today?
questions about your uploaded knowledge base
Summarize Key Documents
Get a high-level overview
Find Latest Updates
Track recent changes
Extract Insights
Surface hidden patterns
Microsoft-Policymaker-Guide-Priva
2 msgs 22/3/2026
SMB_University_120307_Networking
2 msgs 22/3/2026
Message Smart RAG Assistant...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Microsoft-Policymaker-Guide-Priva
2 msgs 22/3/2026
SMB_University_120307_Networking
2 msgs 22/3/2026
SMB_University_120307_Networking
2 msgs 22/3/2026
User
Free Plan
Smart RAG Assistant
Processing gokulganesh_mini_project_paper.docx...
what is in the document
✓ gokulganesh mini_project_paper.docx uploaded
Document uploaded. Processing...
Smart RAG can make mistakes. Verify important information.

FEASIBILITY STUDY

- Evaluates the practicality and viability of the proposed system
- Identifies strengths, weaknesses, opportunities, and required resources

Legal & Ethical Feasibility:

- Ensures data privacy and security
- Complies with regulations (GDPR, NIST)
- Prevents misuse of sensitive information

Technical Feasibility:

- Ensures system can be implemented efficiently
- Uses available technologies and tools

SYSTEM TESTING

- Testing phase ensures reliability, accuracy, and performance of the system
- Validates functionality and identifies errors before deployment

Testing Types:

- **Unit Testing:** Tests individual modules
- **Integration Testing:** Ensures modules work together
- **Functional Testing:** Verifies system functions as expected
- **Performance Testing:** Checks speed and efficiency
- **Security Testing:** Ensures data safety and protection
- **Validation Testing:** Confirms system meets requirements

FUTURE ENHANCEMENTS

- **Cloud Integration:**

  Integrate with AWS, Azure, or Google Cloud for scalable data handling

- **Collaboration:**

  Use real-world network data from organizations to improve accuracy

- **Scalability:**

  Enhance system to handle large-scale networks (banking, healthcare, telecom)

- **Visualization:**

  Add advanced charts, heatmaps, and predictive analytics for better insights

THANK YOU!